

6. Acquisition B: edge-target and Z-step series

The edge series gives lateral (H, V) resolution and most of the boundary-bias data. The Z-step series gives depth resolution and, with the tilted-plate ramp, the sensor's depth quantization. All of it uses robot motion: lateral sub-pixel offsets for edges, small Z steps for depth, and a small tilt for the ramp.

6.1 Edge series (B-HV, also feeds E)

1. Mount T3a, the raised square, with the small gap of 15 mm (GAP_SMALL_MM) and do the mount check of Step 4.8. Rotate it in its own plane by 5° (EDGE_SLANT_DEG), so all four edges are slanted relative to the pixel grid; the as-built record carries the measured slant and the measured gap of each spacer set, and the analyses use those values. The two near-vertical edges measure H resolution and the two near-horizontal edges measure V. Left and right edges have opposite occlusion geometry relative to the baseline, and so do top and bottom.
2. At each shape station, every second station of the ladder (400, 566, 800, 1131, 1600 mm; Z_SHAPE_STATION_STRIDE), centered and fronto-parallel: move, settle, and capture 30 frames (FRAMES_PER_EDGE_POSE) at the nominal pose. The plan tool checks at every shape station that the square is large enough for the analysis bands and the random offsets; at 1600 mm it is 69 px wide at the indicative geometry.
3. At the same station, capture 25 further poses (PHASE_JITTER_POSES_EDGE). Each adds a logged random lateral offset, drawn uniformly over ± 4 px, half the 8 px jitter span (PHASE_JITTER_SPAN_PX), in both H and V, with 30 frames at each. These poses sample every sub-pixel phase of each edge for the robot-stepped edge spread function and for E.
4. Repeat Steps 2–3 with the large gap of 60 mm (GAP_LARGE_MM). Comparing the two step heights tests whether the normalized edge spread function depends on step height. If it does, the depth pipeline is nonlinear and resolution must be quoted together with its step height.
5. Repeat Steps 1–4 with T3b, the square window. The window has the opposite edge polarity: front plate outside, back plate inside.

6.2 Z-step and ramp series (B-Z)

1. Mount T2 and do the mount check. Visit each Z_0 of the reduced stations, every fourth station of the ladder (400, 800, 1600 mm; Z_REDUCED_STATION_STRIDE), centered and fronto-parallel. Approach every visit from below: overshoot by 2 mm toward the sensor (Z_STEP_APPROACH_OVERSHOOT_MM) and move up onto the pose, so that backlash and

compliance do not enter the difference between visits. The read-back robot pose, carried into the camera frame through the registration, is the step truth; log it at 0.01 mm or finer (POSE_LOG_RESOLUTION_MM).

2. Step ladder. The rungs are multiples of the expected depth quantum at the station: 0.25, 0.5, 1, 2, 4, and 8 quanta (Z_STEP_LADDER_QUANTA), using the Tier-A q until A has measured one. A rung below the smallest Z move the robot is trusted to make, 0.1 mm (ROBOT_MIN_RESOLVABLE_MOVE_MM), is raised to it. At the indicative geometry the rungs run from 0.1 to 3.1 mm at 400 mm, 0.39 to 12.4 mm at 800 mm, and 1.55 to 50 mm at 1600 mm; the plan tool lists them. For each rung δ , alternate the target between Z_0 and $Z_0 + \delta$ for 10 cycles (Z_STEP_REPEATS; A, B, A, B, ...), with 10 frames at each visit (FRAMES_PER_ZSTEP_POSE). Log the read-back pose at every visit; the analysis uses the read-back displacement, not the commanded δ , as the step truth, and reports the ratio of the robot repeatability, 0.05 mm (ROBOT_REPEATABILITY_MM), to each rung as the truth uncertainty.
3. Ramp. At every station of the ladder, tilt T2 about H by the angle that makes the true depth across the plate's visible height span 4 expected quanta (RAMP_QUANTA), and capture 50 frames (FRAMES_PER_RAMP_POSE) with the read-back pose logged. Each image row then lies at one true depth, and the rows step through the quanta: at the indicative geometry the tilt is 0.3° at 400 mm and 3.6° at 1600 mm, with 40 to 120 rows per quantum. One capture per station replaces a staircase of small Z moves and does not depend on the robot resolving them. Section 11.2 reads the quantum from the plateaus.
4. Blank windows. No extra capture is needed: the Z_0 frames from Step 2 serve as the no-step reference for the B-Z false-alarm threshold.
5. Optional second pass, the staircase. Where the ramp has measured a quantum of at least 1 mm, the staircase can confirm it in time at one pixel: sweep from Z_0 to $Z_0 + 3$ quanta (Z_STAIRCASE_QUANTA) in steps of one tenth of a quantum (Z_STAIRCASE_SUBDIVISION), never below 0.1 mm (ROBOT_MIN_RESOLVABLE_MOVE_MM), with 10 frames per step (Z_STAIRCASE_FRAMES) and the read-back pose logged. It is outside the Section 9 budget and shows hysteresis or a temporal filter's response to motion, which the static ramp cannot.